Do Consumers Prefer Creator Authenticity Over Brand Authenticity?

1. Introduction

In recent years, influencer marketing has become one of the most dominant forms of digital advertising. Brands increasingly collaborate with social media creators to promote products in a way that appears more personal, relatable, and trustworthy than traditional advertisements. Despite its rapid growth, however, advertising performance in influencer marketing remains highly inconsistent. Some sponsored posts generate exceptionally high engagement, while others underperform even when promoted by large creators or well-known brands.

This raises an important question:

> **What actually determines engagement in influencer advertising?**

Existing marketing research often assumes that successful advertising depends on **brand alignment**, such as category fit between the creator and the brand, or semantic consistency between the creator's identity and brand messaging. Under this view, advertisements should perform better when creators promote products that closely match their niche, values, or communication style.

However, social media platforms such as Instagram are highly visual and creator-driven environments. Consumers may not primarily respond to how well a creator matches a brand. Instead, they may respond more strongly to whether the sponsored content still feels authentic to the creator's usual identity.

To investigate this hypothesis, we analyzed a large-scale Instagram influencer dataset containing over **1.6 million posts**, along with creator profiles, brand profiles, captions, and image data. Using engagement metrics, semantic similarity models, and CLIP-based visual representation learning, we examine whether advertising effectiveness is better explained by **brand authenticity** or **creator authenticity**.

Our findings suggest that consumers respond more strongly to **creator authenticity than brand authenticity**. In particular, advertisements that visually deviate from a creator's established style tend to receive significantly lower engagement, even after controlling for audience size.

2. Dataset

2. Dataset and Preprocessing

To investigate the determinants of advertising effectiveness, we used a large-scale Instagram influencer dataset consisting of post metadata, creator profiles, brand profiles, and image files.

The original dataset contained:

1,601,074 posts

Each post included:

- Creator username
- Sponsorship label (organic or sponsored)
- JSON metadata file
- Associated image files

2.1 Image Matching

Since our analysis required visual content, we first identified posts with at least one available image in the image archive files.

After image matching, the number of usable posts was:

417,788 posts

These matched posts were used as the base dataset for further analysis.

2.2 Creator Filtering

To ensure reliable comparison between organic and sponsored content, we only retained creators who had sufficient posting history in both categories.

A creator was considered qualified if:

- Organic posts ≥ 3
- Sponsored posts ≥ 3

After applying this filter:

6,387 creators

155,688 posts

remained.

2.3 Balanced Sampling

To prevent class imbalance between organic and sponsored posts, we performed user-level balanced sampling.

For each creator, we sampled:

- Equal number of organic posts
- Equal number of sponsored posts

This produced:

70,254 balanced posts

2.4 CLIP Subsampling

Since visual embedding extraction is computationally expensive, we created a subset for image representation learning.

Final CLIP sample:

8,000 posts

Composition:

- Organic posts: 4,038
- Sponsored posts: 3,962

These posts were used for CLIP-based visual feature extraction and style consistency analysis.

3. Methodology

To understand what drives engagement in influencer advertising, we designed a multi-stage analysis framework combining engagement metrics, semantic analysis, and visual representation learning.

Our core objective was to determine whether advertising performance is better explained by **brand alignment** or **creator authenticity**.

3.1 Engagement Measurement

To compare creators with different audience sizes, raw likes or comments alone were insufficient.

Therefore, we defined engagement rate (ER) as:

$$ER = \frac{Likes + Comments}{Followers}$$

This normalization allowed fair comparison between creators with small and large follower bases.

In addition, because follower counts were highly skewed, we applied logarithmic transformation:

$$LogFollowers = \log(Followers)$$

This variable was later used as a control factor in regression analysis.

3.2 Visual Representation Learning

To quantitatively analyze visual consistency, we used the CLIP (Contrastive Language–Image Pretraining) model.

Each image was converted into a:

512-dimensional embedding vector

using CLIP image encoder.

All vectors were L2-normalized before similarity calculation.

3.3 Creator Style Baseline

For each creator, we first constructed a personal visual baseline using their organic (non-sponsored) posts.

Let:

$$v_{organic}$$

denote the average embedding vector of a creator's organic posts.

For each sponsored post, let:

$$v_{ad}$$

denote the embedding vector of the advertisement image.

3.4 Style Fit

We define visual style consistency as:

$$StyleFit = \cos(v_{organic}, v_{ad})$$

A higher value indicates that the advertisement visually resembles the creator's usual content.

3.5 Visual Novelty

To quantify deviation from the creator's usual style, we define:

$$Novelty = 1 - StyleFit$$

A higher novelty score indicates that the advertisement visually deviates more from the creator's established identity.

3.6 Statistical Modeling

To examine the effect of novelty on engagement while controlling for creator size, we used ordinary least squares (OLS) regression:

$$ER = \beta_0 + \beta_1 Novelty + \beta_2 \log(Followers) + \epsilon$$

Where:

- β_1: effect of visual novelty
- β_2: control effect of audience size

This model allowed us to isolate whether creator style disruption independently affects advertising performance.

4. Experimental Results

We conducted multiple experiments to determine whether engagement in influencer advertising is better explained by brand alignment or creator authenticity.

4.1 Sponsored Posts vs Organic Posts

We first compared the engagement rates of sponsored posts and organic posts.

Results showed:

Organic Posts

Mean ER = 0.0440

Sponsored Posts

Mean ER = 0.0413

Statistical testing showed:

$p = 0.0098$

This indicates that sponsored posts generate significantly lower engagement than organic posts.

Overall, sponsored posts showed approximately:

6% lower engagement

than organic content.

4.2 Do Advertisements Break Creator Style?

We then compared style consistency between organic posts and sponsored posts.

Organic Posts

Mean Style Fit = 0.948

Sponsored Posts

Mean Style Fit = 0.660

Statistical testing showed:

$p \approx 0$

This suggests that sponsored posts significantly deviate from the creator's usual visual style.

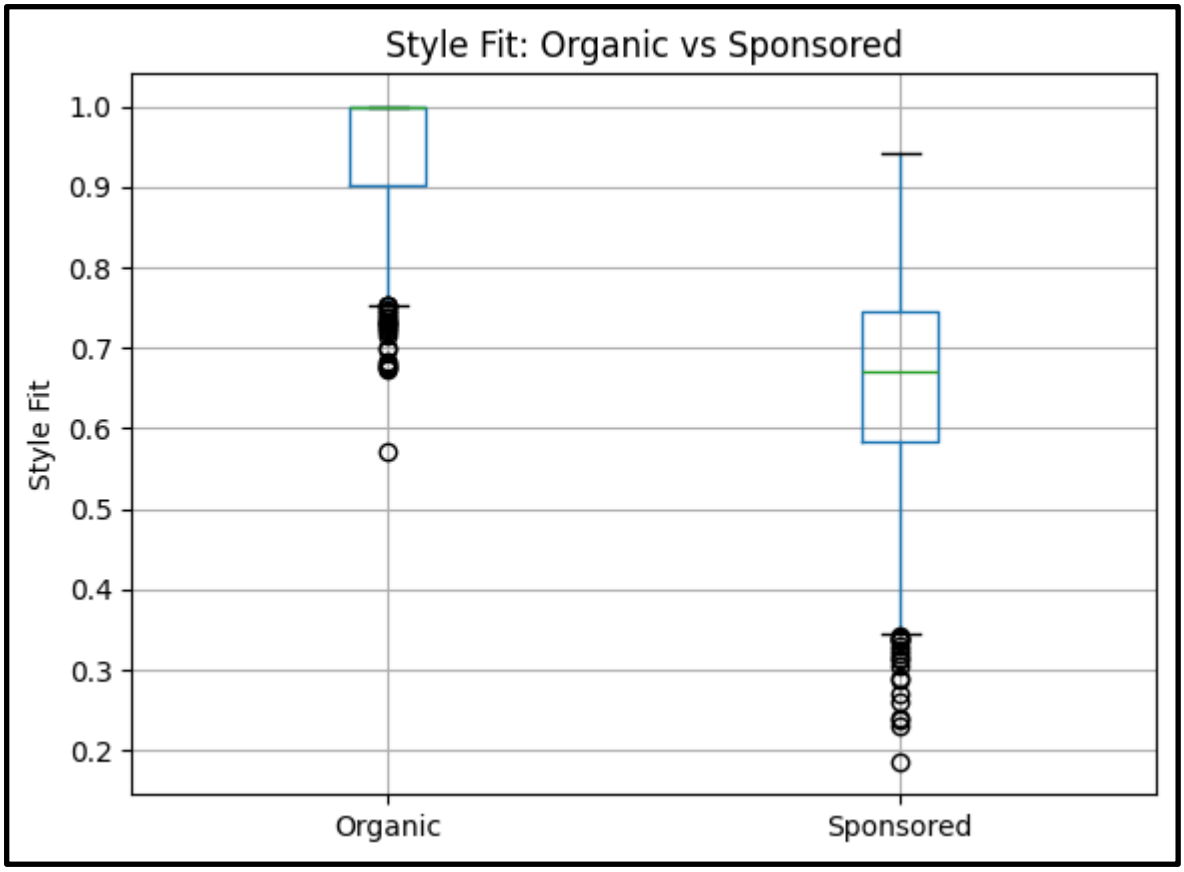

Figure 1. Style fit comparison between organic and sponsored posts. Sponsored posts show substantially lower style fit, indicating that advertisements tend to deviate from creators' usual visual identity.

4.3 Does Visual Novelty Reduce Engagement?

To test whether visual deviation affects advertising performance, we divided advertisements into:

- Low Novelty Ads
- High Novelty Ads

Results:

Low Novelty Ads

Mean ER = 0.0448

High Novelty Ads

Mean ER = 0.0383

This corresponds to approximately:

15% lower engagement

for visually inconsistent advertisements.

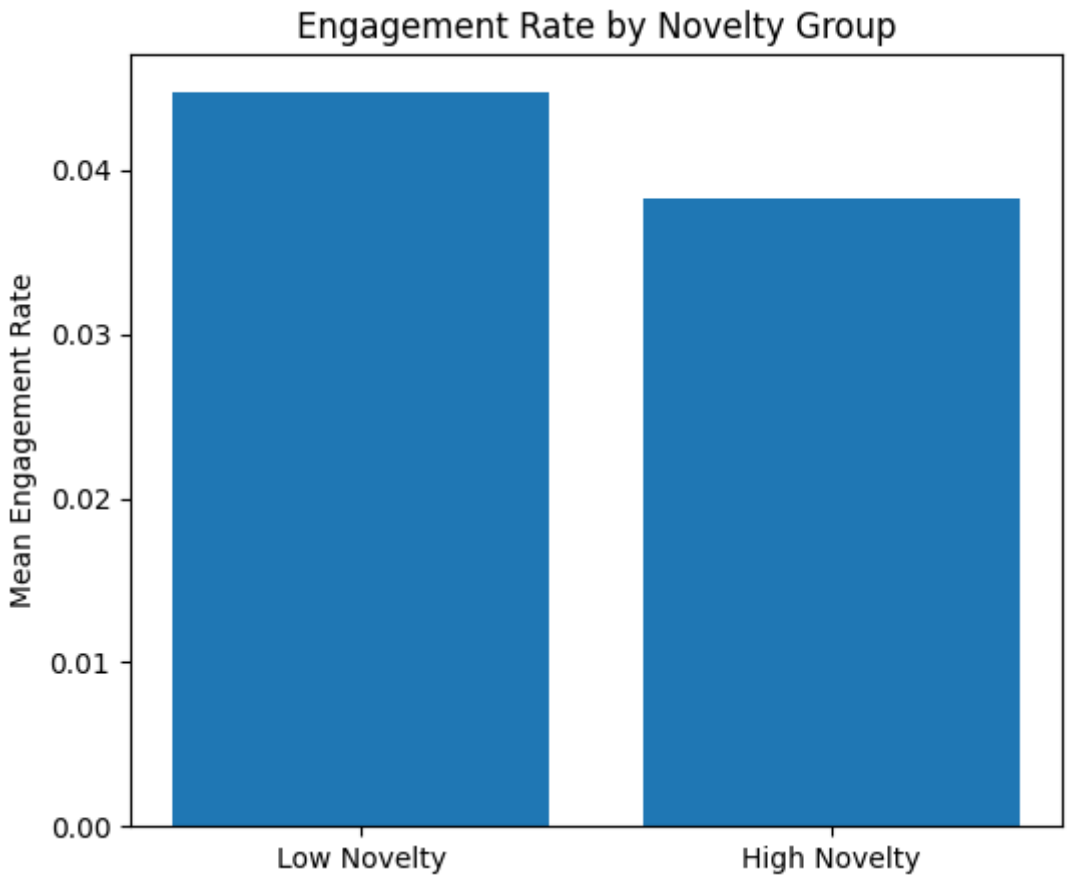

Figure 2. Mean engagement rates of low-novelty and high-novelty advertisements. Advertisements with higher visual novelty receive lower engagement.

4.4 Regression Analysis

To control for follower size, we performed OLS regression.

Results:

Novelty coefficient = -0.0547
$p < 0.001$
$R^2 = 0.118$

This indicates that visual novelty has a statistically significant negative effect on engagement, even after controlling for audience size.

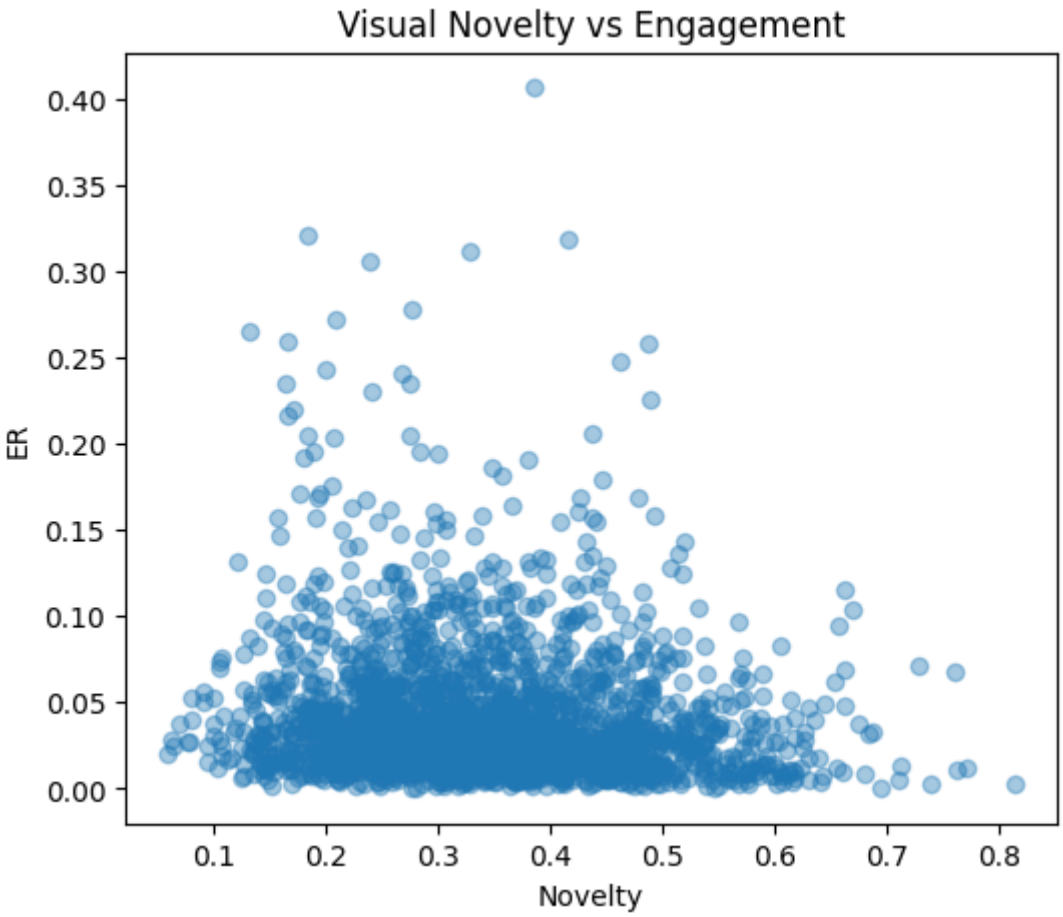

Figure 3. Relationship between visual novelty and engagement rate. Higher visual novelty is associated with lower engagement, supporting the hypothesis that creator-style disruption reduces advertising performance.

4.5 Robustness Checks

To verify that the effect was not driven by outliers, we removed extreme viral posts and repeated the regression.

Results remained significant:

Novelty coefficient = -0.0432
$p < 0.001$

This confirms that the observed effect is robust.

4.6 Category Fit and Semantic Fit

We also tested whether brand alignment explains advertising performance.

Category Fit

Category-matched advertisements did not consistently outperform non-matched advertisements.

Due to limited matched samples, category fit was not considered a reliable predictor.

Semantic Fit

Semantic similarity between advertisement captions and brand descriptions showed:

$p = 0.372$

indicating no statistically significant relationship.

Therefore, textual brand alignment was not found to be a strong determinant of engagement.

5. Discussion and Conclusion

The results of this study challenge several common assumptions in influencer marketing.

Traditional marketing intuition suggests that advertising success depends on **brand alignment**—for example, matching the creator's category with the brand, or ensuring that promotional language closely resembles brand messaging.

However, our empirical results did not consistently support these assumptions.

Neither category fit nor semantic fit showed strong or stable predictive power for engagement.

In contrast, visual authenticity consistently emerged as the strongest predictor of advertising performance.

Our analysis revealed three key findings:

First, sponsored posts significantly deviated from creators' usual visual style.

Second, advertisements with higher visual novelty generated lower engagement.

Third, this effect remained statistically significant even after controlling for follower size, applying log transformations, and removing extreme viral outliers.

These findings suggest that consumers do not primarily respond to how well a creator matches a brand.

Instead, they respond more strongly to whether the advertisement still feels authentic to the creator's established identity.

In other words:

Consumers respond more strongly to creator authenticity than brand authenticity.

From a business perspective, this suggests that brands should not focus solely on selecting creators who fit their product category or brand image.

Instead, brands may achieve better performance by allowing creators to preserve their own visual identity during sponsored content production.

Limitations

This study has several limitations.

First, only a subset of the available image archives was used for visual analysis.

Second, some brand-category matches were limited by incomplete metadata.

Third, this study focused only on Instagram data, and future work should examine whether similar patterns hold across other platforms such as TikTok or YouTube.

Future Work

Future research may extend this work by:

- Incorporating the full image dataset

- Combining visual and textual features in a multimodal model
- Investigating temporal effects such as posting time and trend dynamics

Despite these limitations, the current study provides strong evidence that creator authenticity plays a critical role in influencer advertising effectiveness.